

TRYVANNSHOGDA, Norway

WMO: 014900

Lat: 59.984N

Lon: 10.669E

Elev: 528

StdP: 95.14

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.7	-12.7	-17.2	0.9	-12.7	-15.3	1.1	-11.2	10.7	-2.4	9.5	-2.8	2.0	50	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	23.7	15.3	21.9	14.7	20.0	13.9	16.7	20.9	15.8	19.6	15.0	18.5	2.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.3	11.5	17.6	14.4	10.9	16.8	13.5	10.3	16.1	48.9	21.0	46.1	19.5	43.7	18.6	19.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.7	7.6	6.7	-15.3	25.7	2.4	2.8	-17.0	27.7	-18.4	29.4	-19.8	31.0	-21.6	33.0
			-15.6	17.6	2.4	1.1	-17.3	18.4	-18.7	19.1	-20.1	19.7	-21.9	20.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-4.1	-3.8	-0.7	3.9	8.7	12.6	15.1	13.8	9.8	4.3	0.2	-2.5		
	DBStd	7.69	4.24	4.32	4.34	3.57	4.21	3.20	3.08	2.62	2.72	3.30	3.89	4.40		
	HDD10.0	2324	437	387	332	185	78	8	0	2	37	178	294	386		
	HDD18.3	4952	696	620	591	432	301	177	112	142	257	436	544	644		
	CDD10.0	431	0	0	0	3	37	85	157	119	29	1	0	0		
	CDD18.3	18	0	0	0	0	1	4	11	2	0	0	0	0		
	CDH23.3	62	0	0	0	0	3	13	44	2	0	0	0	0		
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0		
(14) Wind	WSAvg	3.2	3.2	3.0	3.2	3.2	3.2	3.0	2.9	2.9	3.3	3.4	3.4	3.4		
(15) Precipitation	PrecAvg	957	77	58	53	60	63	82	88	109	89	106	102	70		
	PrecMax	1357	187	183	154	117	141	156	181	201	210	257	375	147		
	PrecMin	741	16	17	0	5	3	7	16	24	21	11	22	15		
	PrecStd	147	43	35	34	30	35	39	40	43	44	60	72	34		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	7.1	12.2	15.8	23.2	25.1	26.7	23.3	18.8	12.5	9.5	6.9		
		MCWB	3.1	2.8	6.5	8.6	13.4	14.8	16.7	15.7	13.4	9.7	7.8	5.1		
	2%	DB	4.0	5.0	9.2	13.7	20.1	22.6	24.5	21.6	16.6	10.9	7.8	5.4		
		MCWB	2.3	1.7	4.4	7.7	12.2	13.7	16.0	15.5	12.0	9.2	6.8	4.0		
	5%	DB	2.5	3.2	6.9	11.5	17.8	20.2	22.6	19.8	15.1	9.9	6.5	4.2		
		MCWB	1.2	0.8	2.9	6.4	11.0	12.7	15.6	14.6	11.4	8.7	5.7	3.1		
	10%	DB	1.3	1.6	5.0	9.6	15.5	18.0	20.5	18.0	13.6	8.8	5.3	3.0		
		MCWB	0.3	0.3	1.6	5.1	10.2	11.7	14.8	13.8	10.8	7.6	4.6	1.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.1	7.1	10.1	15.2	16.0	18.2	17.5	14.4	11.4	8.6	5.8		
		MCDB	4.8	6.1	11.4	14.4	19.5	21.4	23.8	20.8	16.7	11.7	8.9	6.2		
	2%	WB	2.6	2.4	4.7	8.4	13.5	14.8	17.2	16.4	13.2	10.1	7.1	4.5		
		MCDB	3.7	4.0	8.4	12.6	18.2	19.9	22.3	19.2	15.1	10.3	7.6	5.1		
	5%	WB	1.4	1.3	3.4	6.9	12.0	13.7	16.2	15.5	12.3	9.0	5.8	3.4		
		MCDB	2.3	2.6	6.1	10.5	16.3	18.5	20.7	18.2	13.8	9.6	6.2	4.0		
	10%	WB	0.3	0.4	2.1	5.7	10.6	12.7	15.4	14.7	11.5	8.0	4.6	2.0		
		MCDB	1.1	1.3	4.2	9.0	14.5	16.8	19.4	17.2	13.0	8.6	5.2	2.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.6	4.0	5.2	6.3	7.1	7.4	6.9	6.1	5.6	4.4	3.6	3.7		
		MCDBR	4.8	5.8	7.6	8.6	9.3	9.6	9.4	7.6	7.4	5.2	4.8	4.8		
	5% WB	MCWBR	4.2	4.3	4.5	4.7	4.8	4.9	4.2	3.6	4.3	4.0	4.3	4.5		
		MCWBR	4.4	4.7	6.9	8.0	8.0	8.1	8.4	6.4	6.0	4.4	4.8	4.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.240	0.280	0.305	0.343	0.341	0.332	0.350	0.344	0.315	0.303	0.257	0.211		
		taud	2.227	2.333	2.385	2.388	2.458	2.498	2.459	2.472	2.545	2.502	2.307	2.118		
	Ebn at Noon	Ebn at Noon	567	739	835	859	888	901	873	851	824	714	540	443		
		Edh at Noon	39	63	82	98	99	96	98	90	71	54	36	27		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.28	0.90	2.38	3.77	4.88	5.59	5.00	3.93	2.60	1.15	0.38	0.19		
	RadStd	RadStd	0.05	0.16	0.25	0.51	0.50	0.42	0.65	0.32	0.30	0.22	0.10	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (58 neighbors)	N/A	N/A	+1.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page